

Prachi Badarayani

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Education

Cornell University

PHD IN COMPUTER SCIENCE

- Advisor: Prof. Volodymyr Kuleshov

New York

Aug 2025

Cornell University

MS IN COMPUTER SCIENCE

- Advisor: Prof. Volodymyr Kuleshov

New York

Aug 2022

Indian Institute of Technology, Bombay

BTECH WITH HONORS IN COMPUTER SCIENCE

- Advisor: Prof. Sudarshan

Mumbai

Aug 2018

Research Interests

My research interests span deep generative probabilistic models, uncertainty estimation and causal machine learning. I work on multimodal generative models for causal inference that incorporate unstructured information for the correction of confounding. I also work on improving the reliability of predictive uncertainties in the context of deep learning, causal inference and sequential decision-making.

Work Experience

MICROSOFT, CAMBRIDGE, USA

JULY 2025 - PRESENT

Dr. Sadid Hasan

Selected for Microsoft's highly selective AI Development Acceleration Program (0.1% acceptance rate), a fixed-term rotational role for new PhDs designed to apply advanced research across multiple AI product teams.

MICROSOFT RESEARCH, REDMOND, USA

MAY 2023 - AUG 2023

Dr. Swati Sharma, Dr. Angels de Luis Balaguer

We built deep generative causal models to power a conversational agent for the FarmBeats project, improving agricultural yields by providing farmers with actionable causal insights for decision-making.

AMAZON ROBOTICS, NORTH READING, USA

JUNE 2022-AUG 2022

Dr. Chaitanya Mitash

We designed vision transformers for complex spatial reasoning in cluttered scenes, enhancing robotic scene understanding to enable the manipulation of previously unseen objects.

NOMURA STRUCTURED FINANCE, MUMBAI, INDIA

JUNE 2017-AUG 2017

Alok Kacchap

Architected and implemented a suite of C++ libraries for the bond futures and options market, engineered for robustness and accuracy against dynamic shifts in market conventions.

SWISS FEDERAL INSTITUTE OF TECHNOLOGY (EPFL), SWITZERLAND

JUNE 2016-AUG 2016

Prof. Bernard Moret

We inferred the evolutionary history of plant genomes using sequential genetic data using maximum likelihood estimation. We demonstrated significant reduction in the computational requirements of phylogenetic (evolutionary) tree reconstruction for plant species using improved representation of sequence data.

Publications

Towards xApp Conflict Evaluation with Explainable Machine Learning and Causal Inference in O-RAN.

Pragya Sharma, Shihua Sun, **Shachi Deshpande**, Angelos Stavrou, Haining Wang
Network Function Virtualization and Software Defined Networks (NFV-SDN), 2025 (**Best Paper Award**)

Calibrated Regression Against An Adversary Without Regret.

Shachi Deshpande, Charles Marx and Volodymyr Kuleshov
Uncertainty in Artificial Intelligence (UAI), 2025

Calibrated and Conformal Propensity Scores for Causal Effect Estimation.

Shachi Deshpande and Volodymyr Kuleshov.
Uncertainty in Artificial Intelligence (UAI), 2024
Also appeared at *Spurious Correlations, Invariance and Stability (SCIS) Workshop, ICML 2023*

Online Calibrated and Conformal Prediction Improves Bayesian Optimization.

Shachi Deshpande, Charles Marx and Volodymyr Kuleshov.
International Conference on Artificial Intelligence and Statistics (AISTATS) 2024

Deep Multi-Modal Structural Equations For Causal Effect Estimation With Unstructured Proxies.

Shachi Deshpande, Kaiwen Wang, Dhruv Sreenivas, Zheng Li and Volodymyr Kuleshov.
Advances in Neural Information Processing Systems (NeurIPS) 2022
Also appeared at *Women in Machine Learning Workshop (WiML)*, *NeurIPS* 2022

Calibrated and Sharp Uncertainties in Deep Learning via Simple Density Estimation.

Volodymyr Kuleshov and **Shachi Deshpande**.
International Conference on Machine Learning (ICML) 2022. (Spotlight)
Also appeared at *Distribution-Free Uncertainty Quantification (DFUQ) Workshop, ICML 2022*.

New Genome Similarity Measures based on Conserved Gene Adjacencies.

Daniel Doerr, Luis Antonio B. Kowada, Eloí Araujo, **Shachi Deshpande**, Simone Dantas, Bernard M.E. Moret, and Jens Stoye. *Journal of Computational Biology* 2017.

Research

CALIBRATION OF LEARNED PROPENSITY SCORE MODELS FOR CAUSAL MACHINE LEARNING

Prof. Volodymyr Kuleshov | Cornell University

We propose probabilistic calibration to improve the effectiveness of propensity score models in causal effect estimation. Calibration reduces the variance and error bounds on causal effect estimates, thus allowing the use of simple propensity models that improve the computational throughput of Genome-Wide Association Studies by more than two-fold.

DEEP MULTIMODAL STRUCTURAL EQUATIONS FOR CAUSAL INFERENCE

Prof. Volodymyr Kuleshov | Cornell University

We incorporate multi-modal, unstructured information in modern datasets within the framework of causal inference using deep generative models. We propose novel generative architectures and inference algorithms that scale to multi-modal setups with missing data. We create multimodal causal inference benchmarks for evaluation of causal effect estimation.

CALIBRATED AND CONFORMAL PREDICTION FOR SEQUENTIAL DECISION-MAKING UNDER UNCERTAINTY

Prof. Volodymyr Kuleshov | Cornell University

We propose a simple algorithm to calibrate the uncertainty of posterior distributions over the objective function as part of the Bayesian optimization process. We show that by improving the uncertainty estimates of the posterior distribution with calibration, Bayesian optimization makes better decisions and arrives at the global optimum in fewer steps. **ONLINE**

QUERY OPTIMIZATION IN DISTRIBUTED SYSTEMS

Prof. S Sudarshan | Undergraduate Thesis, IIT Bombay

We proposed dynamic tuple routing policies to optimize join computation in streaming query application within a distributed computing environment. We designed parallel query execution policies and demonstrated reduction in latency of computation without compromising the application throughput in simulated star join datasets.

DERIVATIVE CLOUDS FOR COMPUTE-INTENSIVE APPLICATIONS

Prof. Umesh Bellur | R & D Project, IIT Bombay

We performed an empirical study of CPU and memory overcommitment on application performance over a range of derivative setups and virtualization frameworks. We determined efficient Virtual Machine cluster configurations corresponding to application requirements through empirical validation of our models.

Awards and Fellowships

2025	Selected in the Microsoft AI Development Acceleration Program , Microsoft
2022	Doctoral Grant for Grace Hopper Celebration , Cornell University
2022	NeurIPS Travel Grant , Women in ML
2018	Cornell University Fellowship , Cornell University
2013	INSPIRE Scholarship , Dept of Science and Technology, Govt of India
2013	Dhirubhai Ambani Scholarship , Reliance Foundation
2008	National Talent Search Examination Scholarship , NCERT, Govt of India

Seminars and Talks

Oct 2023	Causality Discussion Group , TU Darmstadt, Germany (Virtual)
Oct 2023	Microsoft Research , Cambridge, UK (Virtual)
Sept 2022	Machine Learning Reading Group , Cornell Tech, NY, USA
Aug 2022	Amazon Robotics , Boston, MA, USA

Teaching Experience

Spring 2021	Deep Probabilistic and Generative Models , Teaching Assistant
Fall 2020	Applied Machine Learning , Teaching Assistant
Summer 2020	Introduction to Programming Using Python , Teaching Assistant
Spring 2020	Interactive Computer Graphics , Teaching Assistant
Fall 2017,	Computer Programming and Utilization , Teaching Assistant
Spring 2018	

Key Coursework

Causal Machine Learning, Reinforcement Learning, Decision Theory, Topics in Machine Learning and Natural Language Processing (Seminars), Emerging Cloud Technologies, Model Checking, Computer Graphics, Information Retrieval, Probability Theory, Stochastic Processes

Outreach & Professional Development

SERVICE AND OUTREACH

2021-2022	Computer Science Graduate Organization, Cornell Tech , Vice-President
2021-2022	Dept of Computer Science, Cornell , Mentor for incoming Graduate students
2019-2020	CS Graduate Admissions Committee, Cornell , Reviewer
2017-2018	Dept of Computer Science, IIT Bombay , Academic mentor

REVIEWER

NeurIPS 2021, 2022, 2023, 2024
AISTATS 2022, 2023, 2026
ICML 2023
ICLR 2026